NEXTPATH

NEXTPATH

Final Product & Technical Blueprint

IMPACT 2026 HACKATHON

Team Working Document

Opportunity intelligence & application-readiness platform
for Egyptian and global students

NEXTPATH

Final Product + Technical Blueprint

IMPACT 2026 Hackathon

What this document is

This is the master blueprint for NEXTPATH, the opportunity-intelligence and application-readiness platform your team will build during the IMPACT 2026 hackathon. It is the single source of truth for product scope, system architecture, database design, AI layer rules, team responsibilities, development phases, and the demo script.

How to use it

- Read the whole document at least once before splitting work.
- The 87 numbered sections are also your task checklist.
- Part I–II define *what* you are building. Part III–IV define *how the intelligence works*. Part V–VII define *what you actually ship*. Parts VIII–X define *who builds what, when, and what to cut*. Parts XI–XII define *how you win*.

Conventions

- ASCII diagrams live inside code blocks. Render them as-is.
- Hard requirements block eligibility; soft requirements shape the match score.
- AI explains and assists — it never decides eligibility.
- Database + rules = source of truth. AI = interpretation layer.

Table of Contents

PART I — PRODUCT & PROBLEM

- **1.** PRODUCT VISION
- **2.** THE CORE PROBLEM
- **3.** THE GOLDEN RULE
- **4.** COMPLETE SYSTEM ARCHITECTURE
- **5.** STUDENT EXPERIENCE
- **6.** STEP 2 — PROFILE
- **7.** PROFILE COMPLETENESS
- **8.** OPPORTUNITY MODEL
- **9.** OPPORTUNITY DATABASE
- **10.** OPPORTUNITY DATA PIPELINE
- **11.** IMPORTANT: DON'T SCRAPE THE ENTIRE INTERNET

PART II — OPPORTUNITY DATA & ELIGIBILITY

- **12.** VERIFICATION SYSTEM
- **13.** ELIGIBILITY ENGINE
- **14.** HARD REQUIREMENTS
- **15.** SOFT REQUIREMENTS
- **16.** FOUR ELIGIBILITY STATES

PART III — MATCHING & AI LAYER

- **17.** MATCH ENGINE
- **18.** MATCH SCORE
- **19.** WHY THE MATCH?
- **20.** “WHY NOT?” FEATURE
- **21.** “WHAT AM I MISSING?”
- **22.** AI NATURAL LANGUAGE SEARCH
- **23.** AI MUST NOT SEARCH THE INTERNET AND MAKE UP RESULTS
- **24.** AI SERVICE ARCHITECTURE
- **25.** AI APPLICATION ASSISTANT

PART IV — APPLICATIONS

- **26.** APPLICATION PROFILE
- **27.** APPLICATION FIELD MAPPING
- **28.** APPLICATION ASSISTANT — MVP VERSION
- **29.** AUTO-APPLY
- **30.** FUTURE AUTO-APPLY ARCHITECTURE

PART V — OPPORTUNITY TAXONOMIES

- **31.** VOLUNTEERING
- **32.** EVENTS
- **33.** GLOBAL OPPORTUNITIES
- **34.** LOCATION MODEL
- **35.** FUNDING MODEL
- **36.** DEADLINE ENGINE
- **37.** APPLICATION TRACKER
- **38.** NOTIFICATIONS

PART VI — PRODUCT SURFACE (UI / PAGES)

- **39.** DASHBOARD
- **40.** DISCOVERY PAGE
- **41.** AI SEARCH PAGE
- **42.** OPPORTUNITY DETAIL PAGE

PART VII — DATABASE, TECH STACK & API

- **43.** DATABASE / ERD
- **44.** DATABASE TABLES
- **45.** TECH STACK
- **46.** DATABASE
- **47.** AI
- **48.** SEARCH
- **49.** API STRUCTURE

PART VIII — SECURITY, TEAM & GITHUB

- **50.** SECURITY
- **51.** AI PRIVACY
- **52.** TEAM STRUCTURE
- **53.** PERSON 2 — BACKEND / DATABASE
- **54.** PERSON 3 — AI / DATA
- **55.** TEAM DEPENDENCIES
- **56.** GITHUB STRUCTURE

PART IX — DEVELOPMENT PHASES

- **57.** DEVELOPMENT PHASE 1
- **58.** DEVELOPMENT PHASE 2
- **59.** DEVELOPMENT PHASE 3
- **60.** DEVELOPMENT PHASE 4
- **61.** DEVELOPMENT PHASE 5
- **62.** DEVELOPMENT PHASE 6
- **63.** DEVELOPMENT PHASE 7

- **64.** DEVELOPMENT PHASE 8

PART X — SCOPE (MVP, SECONDARY, FUTURE, DROP)

- **65.** MVP DEFINITION
- **66.** SECONDARY FEATURES
- **67.** FUTURE FEATURES
- **68.** FEATURES I WOULD DROP FOR NOW

PART XI — POSITIONING & STRATEGY

- **69.** COMPETITOR LESSONS
- **70.** YOUR DIFFERENTIATION
- **71.** THE “EGYPTIAN LENS”
- **72.** OPPORTUNITY INTELLIGENCE
- **73.** APPLICATION PRIORITY
- **74.** “OPPORTUNITY RADAR”
- **75.** PROFILE GAP ANALYSIS
- **76.** LONG-TERM PRODUCT
- **77.** THE COMPLETE USER FLOW

PART XII — HACKATHON DEMO & FINAL PITCH

- **78.** HACKATHON DEMO
- **79.** Scene 2 — Create student
- **80.** Scene 3 — AI SEARCH
- **81.** Scene 4 — KILLER MOMENT
- **82.** Scene 5 — WHY NOT
- **83.** Scene 6 — APPLICATION
- **84.** Scene 7 — TRACKING
- **85.** FINAL DEMO STATEMENT
- **86.** FINAL FEATURE PRIORITY MATRIX
- **87.** THE FINAL ARCHITECTURE TO BUILD

NEXTPATH

Final Product + Technical Blueprint

IMPACT 2026 Hackathon

1. PRODUCT VISION

```
         NEXTPATH
            │
            ▼
┌───────────────────────────┐
│  STUDENT PROFILE          │
│Education                  │
│Major                      │
│GPA                        │
│Age                        │
│Nationality                │
│Skills                     │
│Languages                  │
│Interests                  │
│Goals                      │
│Experience                 │
└───────────────────────────┘
            │
            ▼
┌───────────────────────────┐
│OPPORTUNITY INTELLIGENCE   │
│                           │
│Scholarships               │
│Internships                │
│Hackathons                 │
│Research                   │
│Exchanges                  │
│Training                   │
│Events                     │
│Volunteering               │
│Competitions               │
└───────────────────────────┘
            │
            ▼
┌───────────────────────────┐
│ELIGIBILITY ENGINE         │
│                           │
│Can they apply?            │
│Why?                       │
│Why not?                   │
│What's missing?            │
└───────────────────────────┘
            │
            ▼
┌───────────────────────────┐
│MATCH + PRIORITIZATION     │
│                           │
│Match Score                │
│Deadline                   │
│Funding                    │
│Personal Goals             │
│Competitiveness            │
└───────────────────────────┘
            │
            ▼
┌───────────────────────────┐
│APPLICATION ASSISTANT      │
│                           │
│Reusable profile           │
│Autofill                   │
│Documents                  │
│AI writing assistance      │
│Application tracking       │
└───────────────────────────┘
            │
            ▼
       STUDENT ACTION
```

2. THE CORE PROBLEM

Students don't necessarily lack opportunities.

They lack **opportunity intelligence**.

```
Facebook
Telegram
WhatsApp
LinkedIn
University websites
Scholarship websites
Company websites
NGOs
Government websites
Random opportunity pages
             │
             ▼
       INFORMATION CHAOS
             │
             ▼
   ┌──────────────────────────┐
   │ Student asks:            │
   │                          │
   │ "Can I apply?"           │
   │ "Am I eligible?"         │
   │ "Is Egypt accepted?"     │
   │ "What do I need?"        │
   │ "When is the deadline?"  │
   │ "Which one is best?"     │
   └──────────────────────────┘
              │
              ▼
       HOURS OF SEARCHING
              │
              ▼
       MISSED OPPORTUNITIES
```

NEXTPATH changes this to:

```
Student
  ↓
Profile
  ↓
"Find opportunities for me"
  ↓
NEXTPATH
  ↓
Eligible opportunities
  ↓
Match score
  ↓
Why you're eligible
  ↓
What you're missing
  ↓
Deadline priority
  ↓
Application assistance
```

3. THE GOLDEN RULE

This should be the most important architectural principle in the entire project:

AI does NOT decide eligibility.

Instead:

```
        OFFICIAL SOURCE
               │
               ▼
    STRUCTURED REQUIREMENTS
               │
               ▼
      RULE-BASED ENGINE
               │
       ┌───────┴────────┐
       ▼                ▼
 ELIGIBILITY         MATCHING
       │                │
       └───────┬────────┘
               ▼
               AI
               │
               ▼
          EXPLANATION
```

Therefore:

Database + rules = source of truth

AI = interpretation, search, explanation and assistance

This protects you from hallucinations.

---

4. COMPLETE SYSTEM ARCHITECTURE

```
                    ┌────────────────────┐
                    │      STUDENT       │
                    └─────────┬──────────┘
                              │
                              ▼
                    ┌────────────────────┐
                    │    NEXT.JS WEB     │
                    │    APPLICATION     │
                    └─────────┬──────────┘
                              │
               ┌──────────────┼──────────────┐
               │              │              │
               ▼              ▼              ▼
           Dashboard        Search         Profile
               │              │              │
               └──────────────┼──────────────┘
                              ▼
                    ┌────────────────────┐
                    │    BACKEND API     │
                    └─────────┬──────────┘
                              │
     ┌────────────────────────┼────────────────────────┐
     │                        │                        │
     ▼                        ▼                        ▼
┌──────────────────┐ ┌──────────────────┐ ┌─────────────────┐
│ Eligibility      │ │ Match / Ranking  │ │ AI Services     │
│ Engine           │ │ Engine           │ │                 │
└────────┬─────────┘ └────────┬─────────┘ └───────┬─────────┘
         │                    │                   │
         └────────────────────┼───────────────────┘
                              ▼
                    ┌────────────────────┐
                    │    PostgreSQL      │
                    │    SUPABASE        │
                    └─────────┬──────────┘
                              │
            ┌─────────────────┼─────────────────┐
            │                 │                 │
            ▼                 ▼                 ▼
       Opportunities       Profiles        Applications
            ▲
            │
   ┌────────────────────┐
   │  OPPORTUNITY DATA  │
   └─────────┬──────────┘
             │
   ┌─────────┼─────────┐
   ▼         ▼         ▼
 Official   APIs     Manual
 Websites            Sources
   │         │         │
   └─────────┼─────────┘
             ▼
       Data Extraction
             │
             ▼
       Normalization
             │
             ▼
       Verification
             │
             ▼
       Opportunity DB
```

5. STUDENT EXPERIENCE

Step 1 — Account

```
Create account
      ↓
Email / OAuth
      ↓
Create profile
```

6. STEP 2 — PROFILE

The profile is the foundation of NEXTPATH.

```
                STUDENT PROFILE
                       │
    ┌──────────────────┬──────────────────┐
    ▼                  ▼                  ▼
 Personal          Education          Academic
    │                  │                  │
 Name               School             GPA
 Age                University          Grades
 Nationality        Degree             Year
 Country            Major
    │
    ├──────────────────────────────────────┐
    ▼                                      ▼
 Skills                                Languages
    │                                      │
 Python                                 English
 C++                                    Arabic
 AI                                     French
 etc.                                   etc.
    │
    ├──────────────────────────────────────┐
    ▼                                      ▼
Experience                              Interests
    │                                      │
Projects                               AI
Internships                            Robotics
Volunteering                           Cybersecurity
    │
    ├──────────────────────────────────────┐
    ▼                                      ▼
  Goals                                Documents
    │                                      │
Scholarship                              CV
Internship                              Certificates
Research                                Transcripts
```

7. PROFILE COMPLETENESS

Do **not** force students to fill everything.

Instead:

```
Your profile is 78% complete

████████████████░░░░

✓ Education
✓ Major
✓ GPA
✓ Nationality
✓ Age
✓ Skills
✓ Languages

⚠ Add:
   CV
   Interests
   Experience

Adding these could unlock
23 more opportunities.
```

This gives the student a reason to complete their profile.

---

8. OPPORTUNITY MODEL

Everything should use one universal model.

```
                    OPPORTUNITY
                         │
        ┌────────────────┼─────────────────┐
        ▼                ▼                 ▼
       TYPE            BASIC             LOCATION
                     INFORMATION
        │                │                  │
 Scholarship        Title              Country
 Internship         Organization       City
 Hackathon           Description        Remote
 Research            Website
 Exchange            Funding
 Training            Deadline
 Event
 Volunteer
 Competition
```

9. OPPORTUNITY DATABASE

Each opportunity should contain:

```
Opportunity
│
├── ID
├── Title
├── Type
├── Organization
├── Description
├── Category
├── Field
│
├── Eligibility
│   ├── Nationality
│   ├── Age
│   ├── Education
│   ├── Academic Year
│   ├── GPA
│   ├── Major
│   ├── Language
│   ├── Skills
│   ├── Experience
│   └── Residency
│
├── Funding
│   ├── Fully Funded
│   ├── Partially Funded
│   ├── Paid
│   └── Unpaid
│
├── Location
├── Remote
├── Start Date
├── End Date
├── Deadline
│
├── Required Documents
├── Application URL
│
└── Verification
    ├── Source
    ├── Last Verified
    └── Verification Status
```

10. OPPORTUNITY DATA PIPELINE

This is one of the most important diagrams.

```
         OFFICIAL SOURCES
                │
    ┌───────────┼────────────┐
    ▼           ▼            ▼
Universities Companies    NGOs
    │           │            │
    ▼           ▼            ▼
 Websites      APIs       Websites
    │           │            │
    └───────────┼────────────┘
                │
                ▼
         DATA COLLECTOR
                │
                ▼
          RAW OPPORTUNITY
                │
                ▼
         DATA EXTRACTION
                │
                ▼
      STRUCTURED OPPORTUNITY
                │
                ▼
          NORMALIZATION
                │
                ▼
       DUPLICATE DETECTION
                │
                ▼
      ELIGIBILITY EXTRACTION
                │
                ▼
          VERIFICATION
                │
         ┌──────┴──────┐
         ▼             ▼
      VERIFIED      REVIEW NEEDED
         │             │
         └──────┬──────┘
                ▼
         OPPORTUNITY DB
```

11. IMPORTANT: DON'T SCRAPE THE ENTIRE INTERNET

This is one of the biggest traps.

You **cannot realistically promise**:

> *"NEXTPATH contains every opportunity in the world."*

Instead:

MVP

Build a curated database from trusted sources.

For example:

```
Scholarships      20+
Internships       20+
Hackathons        15+
Research          10+
Exchange          10+
Volunteer         10+
Events            10+
```

Even **100 excellent opportunities** can make a much better demo than 10,000 unreliable ones.

---

12. VERIFICATION SYSTEM

```
           OPPORTUNITY
                │
                ▼
      Is source official?
                │
        ┌───────┴───────┐
        │               │
       YES              NO
        │               │
        ▼               ▼
   Extract rules     Reject/Review
        │
        ▼
   Check deadline
        │
        ▼
   Check application URL
        │
        ▼
   Check organization
        │
        ▼
     Confidence
        │
    ┌───┴────┐
    ▼        ▼
  HIGH     MEDIUM
    │        │
    ▼        ▼
 VERIFIED   HUMAN REVIEW
```

Each opportunity should show:

> ***Verified from official source***

and:

> ***Last verified: [date]***

---

13. ELIGIBILITY ENGINE

```
            STUDENT PROFILE
                   │
                   │
                   ▼
         ┌────────────────────┐
         │ ELIGIBILITY ENGINE │
         └─────────┬──────────┘
                   │
    ┌──────────────┼───────────────┐
    ▼              ▼               ▼
Nationality       Age          Education
    │              │               │
    ▼              ▼               ▼
  PASS/FAIL      PASS/FAIL       PASS/FAIL
    │              │               │
    └──────────────┼───────────────┘
                   │
    ┌──────────────┼───────────────┐
    ▼              ▼               ▼
   GPA            Major          Language
    │              │               │
    ▼              ▼               ▼
  PASS/FAIL      PASS/FAIL       PASS/FAIL
                   │
                   ▼
            HARD REQUIREMENTS
                   │
            ┌──────┴──────┐
            ▼             ▼
          FAIL           PASS
            │             │
            ▼             ▼
      NOT ELIGIBLE     MATCH ENGINE
```

14. HARD REQUIREMENTS

These determine whether someone can apply.

```
Nationality
Age
Education level
Academic year
GPA minimum
Residency
Required degree
Deadline
Legal requirements
```

If a hard requirement fails:

```
   NOT ELIGIBLE
```

15. SOFT REQUIREMENTS

These affect the match.

```
Major
Skills
Interests
Experience
Career goals
Language strength
Relevant projects
```

These shouldn't necessarily block the student.

16. FOUR ELIGIBILITY STATES

Don't make everything binary.

```
    ELIGIBLE

    LIKELY ELIGIBLE

    UNKNOWN

    NOT ELIGIBLE
```

Example:

```
    Egyptian nationality

    ELIGIBLE

The official opportunity requirements
explicitly allow Egyptian citizens.
```

Unknown example:

```
    Egyptian nationality

    UNKNOWN

The official page says "international applicants"
but does not explicitly list Egypt.

Verify before applying.
```

This is much more trustworthy than pretending AI knows everything.

17. MATCH ENGINE

Separate **eligibility** from **match**.

```
Eligibility
     │
     ▼
Can you apply?
     │
     ▼
YES
     │
     ▼
Match Engine
     │
 ┌───┼───────────────┐
 ▼   ▼               ▼
Major Skills      Interests
 │     │             │
 └─────┼─────────────┘
       ▼
    Experience
       │
       ▼
      Goals
       │
       ▼
    Language
       │
       ▼
  Match Score
```

18. MATCH SCORE

Example:

```
              MATCH SCORE

Major                  20/20
Skills                 18/20
Interests              15/15
Goals                  14/15
Experience              8/10
Language                9/10
Academic                9/10
                       ──────
                       93%
```

> ***93% Match***

---

19. WHY THE MATCH?

Click:

“Why 93%?”

```
✓ Computer Engineering matches
✓ AI interest matches
✓ Python matches
✓ First-year requirement matches
✓ GPA requirement matches

⚠ Research experience preferred

Your profile doesn't currently show
research experience.
```

This makes the AI explainable.

---

20. “WHY NOT?” FEATURE

This should be one of your signature features.

```
           OPPORTUNITY
                │
                ▼
        Eligibility Engine
                │
                ▼
              FAIL
                │
                ▼
           WHY NOT?
                │
    ┌───────────┼────────────┐
    ▼           ▼            ▼
   Age         GPA      Nationality
    │           │            │
    ▼           ▼            ▼
  19          2.9          Egypt
  req. 21     req. 3.2     not accepted
```

> *You cannot apply because this program requires a minimum GPA of 3.2 and your current GPA is 2.9.*

This is **much more useful than just hiding the opportunity**.

21. "WHAT AM I MISSING?"

Another excellent feature.

```
Student
   │
   ▼
Profile
   │
   ▼
Opportunity database
   │
   ▼
Find near-matches
   │
   ▼
Identify missing requirements
   │
   ▼
┌─────────────────────────────┐
│ You are close to 14 more    │
│ opportunities.              │
│                             │
│ Missing:                    │
│                             │
│ IELTS 6.5                   │
│ Research experience         │
│ Recommendation letter       │
└─────────────────────────────┘
```

Priority:

Build this if the core MVP is finished.

It's more valuable than many flashy features.

---

22. AI NATURAL LANGUAGE SEARCH

Student writes:

> *Find fully funded AI summer programs for Egyptian first-year students.*

```
               USER QUERY
                   │
                   ▼
             AI QUERY PARSER
                   │
                   ▼
          STRUCTURED QUERY
                   │
    ┌──────────────┼───────────────┐
    ▼              ▼               ▼
   AI         Fully funded      Summer
  Field         Funding         Program
    │              │               │
    └──────────────┼───────────────┘
                   ▼
             Egypt eligible
                   │
                   ▼
             Database Search
                   │
                   ▼
           Eligibility Engine
                   │
                   ▼
             Match Engine
                   │
                   ▼
                RESULTS
```

23. AI MUST NOT SEARCH THE INTERNET AND MAKE UP RESULTS

Bad architecture:

```
User
  ↓
LLM
  ↓
"Here are 10 scholarships"
```

Good architecture:

```
User
  ↓
LLM understands request
  ↓
Structured search
  ↓
Verified database
  ↓
Eligibility engine
  ↓
Match engine
  ↓
LLM explains
  ↓
Student
```

24. AI SERVICE ARCHITECTURE

```
                        AI LAYER
                           │
          ┌────────────────┼──────────────────┐
          ▼                ▼                  ▼
   Query Parser      Requirement         Explanation
                     Extraction
          │                │                  │
          ▼                ▼                  ▼
  Natural language    Opportunity          Eligibility
   → filters          → structured         → human explanation
                           data
                           │
                           ▼
                    Application AI
                           │
                           ▼
                     Draft assistance
```

25. AI APPLICATION ASSISTANT

```
Student Profile
      │
      ▼
Opportunity
      │
      ▼
Application Question
      │
      ▼
     AI
      │
      ▼
Draft Answer
      │
      ▼
Student Review
      │
      ▼
Edit
      │
      ▼
Submit
```

AI rule:

> *Never invent student achievements, experience, awards, grades, skills or projects.*

---

26. APPLICATION PROFILE

One of the strongest practical features.

```
                    PROFILE
                       │
         ┌─────────────┼──────────────┐
         ▼             ▼              ▼
   Personal       Education       Experience
         │             │              │
         └─────────────┼──────────────┘
                       ▼
                APPLICATION DATA
                       │
                       ▼
              FIELD MAPPING ENGINE
                       │
         ┌─────────────┼──────────────┐
         ▼             ▼              ▼
    Application A  Application B  Application C
```

27. APPLICATION FIELD MAPPING

For example:

```
Student Profile
───────────────

first_name = Ahmed
last_name = Elhennawy
university = MUST
major = Computer Engineering
```

Different websites:

```
Website A
First Name

Website B
Given Name

Website C
Legal First Name
```

NEXTPATH maps:

```
first_name
    ↓
First Name
Given Name
Legal First Name
Forename
```

28. APPLICATION ASSISTANT — MVP VERSION

When student clicks:

> ***Apply with NEXTPATH***

Show:

```
APPLICATION READINESS

24 fields required

20 ✓ Available from your profile
2  ⚠ Need confirmation
2     Missing

────────────────────

✓ Personal information
✓ Education
✓ Skills
✓ Experience
✓ Documents

⚠ Motivation essay
⚠ Availability dates

[Continue]
```

29. AUTO-APPLY

This requires careful scope.

DON'T BUILD UNIVERSAL AUTO-APPLY

Trying to automatically operate every external website is extremely difficult because:

```
Every website has different:
- Forms
- CAPTCHA
- Login
- File upload
- Questions
- Validation
- Anti-bot systems
- Consent requirements
- Application flows
```

Hackathon:

Build profile reuse + application preparation

Later:

Browser-assisted autofill

Much later:

Autonomous browser agent

---

30. FUTURE AUTO-APPLY ARCHITECTURE

Eventually:

```
NEXTPATH
    │
    ▼
Application Agent
    │
    ▼
Open official application
    │
    ▼
Analyze form
    │
    ▼
Map fields
    │
    ▼
Fill profile information
    │
    ▼
Upload documents
    │
    ▼
AI-assisted questions
    │
    ▼
STOP
    │
    ▼
Student reviews
    │
    ▼
Student submits
```

I strongly recommend keeping the **final submit under user control**.

---

31. VOLUNTEERING

YES — INCLUDE IT

But as another opportunity type, not another product.

```
             OPPORTUNITY
                  │
        ┌─────────┼────────────┐
        ▼         ▼            ▼
Scholarship    Internship   Volunteer
        │         │            │
        └─────────┼────────────┘
                  ▼
         SAME ELIGIBILITY ENGINE
```

Volunteer platforms show that users benefit from filtering by destination, dates, category, cost and other practical constraints, while vetting and safety information are also important.

Add volunteer-specific fields:

```
Age
Nationality
Destination
Duration
Remote
Accommodation
Meals
Travel
Visa
Cost
Language
Physical requirements
```

Priority:

MVP if data is available

Otherwise:

Future

---

32. EVENTS

YES — INCLUDE

But don't build a separate event system.

```
Opportunity
        │
        ├── Scholarship
        ├── Internship
        ├── Hackathon
        ├── Research
        ├── Exchange
        ├── Training
        ├── Volunteer
        ├── Competition
        └── Event
```

Events simply have:

```
Event
├── Date
├── Location
├── Online/Offline
├── Registration deadline
├── Cost
├── Eligibility
└── Registration URL
```

33. GLOBAL OPPORTUNITIES

This is important.

Don't limit the database to:

> *“Scholarships in Egypt.”*

Instead:

```
Student nationality:
   Egypt

          ↓

        WORLD
 ┌─────────┼─────────┐
 ▼         ▼         ▼
Europe    USA       Asia
 ▼         ▼         ▼
Africa    Canada    Australia
           │
           ▼
       Egypt Eligibility
           │
           ▼
      Personalized results
```

The key isn't:

> *Where is the opportunity?*

It's:

> ***Can this Egyptian student apply?***

---

34. LOCATION MODEL

Use:

```
location_type

REMOTE
IN_PERSON
HYBRID
```

```
country
city
region
```

This allows:

> *“Find remote internships.”*

or:

> *“Find opportunities in Europe.”*

---

35. FUNDING MODEL

Use standardized categories:

```
FULLY_FUNDED
PARTIALLY_FUNDED
PAID
UNPAID
FREE
FEE_REQUIRED
UNKNOWN
```

For fully funded programs:

```
Tuition
Accommodation
Flights
Visa
Meals
Stipend
Insurance
```

Each should be individually represented where possible.

---

36. DEADLINE ENGINE

```
Opportunity
     │
     ▼
Deadline
     │
     ▼
Deadline Service
     │
 ┌───┼──────────────┐
 ▼   ▼              ▼
30d  7d             2d
 │    │              │
 ▼    ▼              ▼
Reminder
```

```
   Closing Soon

AI Research Program

Deadline:
12 September

   8 days remaining
```

37. APPLICATION TRACKER

```
               MY APPLICATIONS

 ┌──────────────────────────────────┐
 │ AI Research Program              │
 │    In Progress                   │
 │ Deadline: 8 days                 │
 └──────────────────────────────────┘

 ┌──────────────────────────────────┐
 │ Global Hackathon                 │
 │    Submitted                     │
 └──────────────────────────────────┘

 ┌──────────────────────────────────┐
 │ Scholarship X                    │
 │    Missing Documents             │
 └──────────────────────────────────┘
```

```
DISCOVERED
SAVED
PREPARING
READY
SUBMITTED
ACCEPTED
REJECTED
EXPIRED
```

38. NOTIFICATIONS

```
               NOTIFICATION ENGINE
                        │
        ┌───────────────┼────────────────┐
        ▼               ▼                ▼
    Deadline         New Match        Profile
        │               │                │
        ▼               ▼                ▼
  "3 days left"   "94% match"      "Complete CV"
```

For MVP:

- In-app
- Email

Don't build SMS/WhatsApp infrastructure yet.

---

39. DASHBOARD

The dashboard should NOT look like a normal job board.

It should answer:

“What should I do next?”

```
────────────────────────────────────────

Good evening, Ahmed

PROFILE
████████████████░░ 82%

────────────────────────────────────────

   TOP OPPORTUNITIES FOR YOU

94%  AI Summer Research
        Eligible
        Fully funded
        12 days

91%  Global AI Hackathon
        Eligible
        Prize
        8 days

87%  International Internship
        Likely eligible
     ⚠ Missing language information

────────────────────────────────────────

   DEADLINES

2 opportunities closing this week

────────────────────────────────────────

   YOU COULD UNLOCK

14 more opportunities by adding:
✓ IELTS score
✓ CV
✓ Research experience

────────────────────────────────────────
```

40. DISCOVERY PAGE

```
Discover

[    Search opportunities... ]

Filters:

Type
□ Scholarship
□ Internship
□ Hackathon
□ Research
□ Volunteer
□ Event

Eligibility
□ Eligible
□ Likely eligible

Funding
□ Fully funded
□ Paid
□ Free

Location
□ Remote
□ Europe
□ USA
□ Asia
□ Africa

Deadline
□ This week
□ This month
□ Later
```

41. AI SEARCH PAGE

Make this visually different.

```
        What are you looking for?

Find fully funded AI summer programs
for Egyptian first-year students

            [ Find opportunities ]
```

```
I found 8 opportunities matching your profile.

4    Eligible
2    Likely eligible
2    Not eligible
```

42. OPPORTUNITY DETAIL PAGE

```
AI Summer Research Program

MIT

   94% MATCH

   Egyptians eligible

   Fully funded
   USA
   Deadline: 18 days

────────────────────────

ABOUT

...

────────────────────────

WHY YOU MATCH

✓ First-year student
✓ Computer Engineering
✓ GPA requirement
✓ AI interest

────────────────────────

ELIGIBILITY

Nationality       ✓
Age                ✓
Education          ✓
GPA                ✓
Language           ⚠

────────────────────────

FUNDING

Tuition            ✓
Accommodation      ✓
Travel              ✓
Stipend             ✓

────────────────────────

APPLICATION

Documents:
✓ CV
✓ Transcript
⚠ Recommendation

[ APPLY WITH NEXTPATH ]
```

43. DATABASE / ERD

```
USER
 └────── PROFILE
          │
          ├────── EDUCATION
          ├────── SKILLS
          ├────── LANGUAGES
          ├────── EXPERIENCE
          ├────── INTERESTS
          └────── DOCUMENTS

ORGANIZATION
 └────── OPPORTUNITY
          │
          ├────── REQUIREMENTS
          ├────── FUNDING
          ├────── LOCATION
          ├────── SOURCES
          └────── DOCUMENTS

PROFILE
 │
 ├────── SAVED_OPPORTUNITY ───── OPPORTUNITY
 │
 └────── APPLICATION ──────────── OPPORTUNITY
             │
             ├────── APPLICATION_FIELDS
             └────── APPLICATION_DOCUMENTS

PROFILE
 └────── RECOMMENDATION ──────── OPPORTUNITY
```

44. DATABASE TABLES

Users

```
users
```

Profile

```
profiles
```

Education

```
educations
```

Skills

```
skills
profile_skills
```

Languages

```
languages
profile_languages
```

Experience

```
experiences
```

Documents

```
documents
```

Organizations

```
organizations
```

Opportunities

```
opportunities
```

Requirements

```
opportunity_requirements
```

Sources

```
opportunity_sources
```

Saved

```
saved_opportunities
```

Applications

```
applications
```

Application fields

```
application_fields
```

Notifications

```
notifications
```

Recommendations

```
recommendations
```

45. TECH STACK

Frontend

```
Next.js
TypeScript
Tailwind CSS
```

Backend

For a 3-person team:

```
Next.js API
```

or, if your AI/data developer is stronger in Python:

```
Next.js
      ↓
FastAPI
```

Don't introduce unnecessary backend complexity.

---

46. DATABASE

Supabase

```
SUPABASE
│
├── PostgreSQL
├── Authentication
├── Storage
├── Row-Level Security
└── APIs
```

Excellent for your team because it reduces infrastructure work.

---

47. AI

Use an LLM API for:

```
Natural language search
Requirement extraction
Eligibility explanations
Opportunity summaries
Application writing assistance
Profile gap analysis
```

Do NOT use the LLM as your database.

---

48. SEARCH

Start with:

```
PostgreSQL
+
Structured filters
+
Full text search
+
LLM query parsing
```

```
Embeddings
+
Semantic search
```

You don't need a complicated vector infrastructure for the hackathon.

---

49. API STRUCTURE

```
/api
│
├── auth
│
├── profile
│   ├── GET
│   └── PUT
│
├── opportunities
│   ├── GET
│   └── POST
│
├── opportunities/:id
│   └── GET
│
├── opportunities/:id/eligibility
│   └── GET
│
├── search
│   └── POST
│
├── recommendations
│   └── GET
│
├── saved
│   ├── GET
│   ├── POST
│   └── DELETE
│
├── applications
│   ├── GET
│   ├── POST
│   └── PUT
│
├── documents
│
├── notifications
│
└── ai
    ├── search
    ├── explain
    └── application-assist
```

50. SECURITY

This is particularly important because profiles can contain personal information and documents.

```
Student
   │
   ▼
Authentication
   │
   ▼
Authorized session
   │
   ▼
Backend
   │
   ▼
Row-Level Security
   │
   ▼
Student's data only
```

Never:

```
Student A
   ↓
Student B's profile
```

Documents:

Use private storage.

51. AI PRIVACY

Don't send the student's entire profile to the AI every time.

Bad:

```
Entire profile
+
all documents
+
personal information
→ LLM
```

```
Question:
"Write a motivation letter."

Send:

Major
Skills
Projects
Goals
Relevant experience
Opportunity information
```

Don't send irrelevant sensitive information.

52. TEAM STRUCTURE

You have **3 people**, so don't divide the project into 10 teams.

PERSON 1 — FRONTEND / PRODUCT

```
OWNERSHIP

Landing
Authentication UI
Onboarding
Dashboard
Search
Opportunity pages
Profile
Saved
Applications
Responsive UI
```

53. PERSON 2 — BACKEND / DATABASE

```
OWNERSHIP

Supabase
Database
Authentication
API
Profile storage
Opportunity storage
Applications
Saved opportunities
Notifications
Security
```

54. PERSON 3 — AI / DATA

```
OWNERSHIP

Opportunity data
Data extraction
Normalization
Eligibility engine
Match engine
AI search
AI explanations
AI application assistant
```

55. TEAM DEPENDENCIES

```
            PERSON 2
         Backend / DB
               │
      ┌────────┴────────┐
      ▼                 ▼
 PERSON 1          PERSON 3
Frontend            AI/Data
      │                 │
      └────────┬────────┘
               ▼
           NEXTPATH
```

Person 2 must define:

```
Database schema
API contracts
Opportunity JSON structure
Profile JSON structure
```

This prevents everyone from building incompatible pieces.

---

56. GITHUB STRUCTURE

```
nextpath/
│
├── frontend/
│
├── backend/
│
├── ai/
│
├── data/
│
├── database/
│
├── docs/
│   ├── architecture/
│   ├── diagrams/
│   └── api/
│
├── tests/
│
├── README.md
│
└── .env.example
```

57. DEVELOPMENT PHASE 1

FOUNDATION

```
Day 1
  │
  ├── GitHub
  ├── Next.js
  ├── Supabase
  ├── Database
  ├── Authentication
  └── Basic UI
```

```
User can register
        ↓
Login
        ↓
Create profile
```

58. DEVELOPMENT PHASE 2

PROFILE

```
Profile
  ↓
Education
  ↓
Skills
  ↓
Languages
  ↓
Interests
  ↓
Goals
  ↓
Documents
```

> *Complete student profile.*

---

59. DEVELOPMENT PHASE 3

OPPORTUNITY DATABASE

```
Sources
  ↓
Opportunity records
  ↓
Structured requirements
  ↓
Verification
  ↓
Database
```

Start with a curated dataset.

60. DEVELOPMENT PHASE 4

ELIGIBILITY

```
Profile
+
Opportunity
  ↓
Eligibility engine
  ↓
Eligible / Likely / Unknown / Not eligible
```

This is the first major intelligence milestone.

---

61. DEVELOPMENT PHASE 5

MATCHING

```
Eligible opportunities
        ↓
Major
Skills
Interests
Experience
Goals
Language
        ↓
Match score
        ↓
Ranking
```

62. DEVELOPMENT PHASE 6

AI SEARCH

```
Natural language
       ↓
LLM
       ↓
Structured query
       ↓
Database
       ↓
Eligibility
       ↓
Matching
       ↓
Results
```

63. DEVELOPMENT PHASE 7

APPLICATION ASSIST

```
Opportunity
  ↓
Required fields
  ↓
Profile mapping
  ↓
Missing fields
  ↓
AI assistance
  ↓
Student review
  ↓
External application
```

64. DEVELOPMENT PHASE 8

POLISH

Focus on:

```
UI
Loading states
Error handling
Mobile
Animations
Empty states
Verification badges
Deadline indicators
Demo data
```

Don't add huge features at this point.

---

65. MVP DEFINITION

If you have to cut everything else, keep these:

```
ACCOUNT
STUDENT PROFILE
OPPORTUNITY DATABASE
GLOBAL OPPORTUNITIES
EGYPTIAN ELIGIBILITY
MATCH SCORE
WHY ELIGIBLE?
WHY NOT?
AI NATURAL-LANGUAGE SEARCH
DEADLINES
VERIFIED SOURCES
SAVE
APPLICATION TRACKER
```

That is already a very strong product.

66. SECONDARY FEATURES

Build only after the above works:

```
Volunteer
Events
AI application assistant
Application field autofill
Profile gap analysis
Opportunity alerts
```

67. FUTURE FEATURES

```
Browser-assisted autofill
Autonomous application agent
Skill roadmap
Opportunity forecasting
University dashboards
Organization dashboards
Student communities
Friends
Team formation
Mentors
```

68. FEATURES I WOULD DROP FOR NOW

Social network

Don't build:

```
Friends
Posts
Likes
Comments
Messaging
```

It distracts from the core problem.

Native mobile app

Build responsive web first.

Microservices

You are three people.

Don't spend the hackathon managing:

```
10 services
Docker
Kubernetes
service discovery
message queues
```

69. COMPETITOR LESSONS

Current platforms prove several individual pieces of your idea:

ScholarshipOwl

Uses student profile data to match students with eligible scholarships and supports reusable application information and application tracking.

Circlebreaker

Already offers a global opportunity database, profile matching, decoded eligibility, official links, saved opportunities and deadline-oriented discovery.

Volunteer World

Shows the value of structured international-volunteer search, comparisons, practical filters and quality assurance.

Therefore:

Don't pitch:

> *“We use AI to find scholarships.”*

That's not enough.

Pitch:

> ***“NEXTPATH brings opportunity discovery, verified eligibility, personalized prioritization and application readiness into one intelligence layer designed around the student's actual profile.”***

70. YOUR DIFFERENTIATION

```
Existing platforms:

Opportunity
     ↓
Search
     ↓
Apply

NEXTPATH:

Student
  ↓
Personal profile
  ↓
Global opportunity universe
  ↓
Verified requirements
  ↓
Egyptian eligibility
  ↓
Personal match
  ↓
WHY?
  ↓
WHAT AM I MISSING?
  ↓
WHAT SHOULD I APPLY FOR FIRST?
  ↓
APPLICATION READINESS
  ↓
TRACK
```

71. THE “EGYPTIAN LENS”

This should be a major part of your identity.

For every opportunity:

```
   CAN EGYPTIANS APPLY?

   YES
Official source explicitly accepts Egypt.
```

```
   UNCLEAR

International applicants are accepted,
but Egypt isn't explicitly listed.
```

```
   NO

Eligibility is limited to EU citizens.
```

This makes NEXTPATH immediately useful to your target audience.

72. OPPORTUNITY INTELLIGENCE

Eventually your platform can calculate:

```
                  OPPORTUNITY INTELLIGENCE

Eligibility             100%
Personal Match            93%
Funding                   100%
Deadline Urgency           82%
Profile Readiness          75%

────────────────────────────

OVERALL PRIORITY

   HIGH PRIORITY
```

Don't call this “chance of acceptance.”

You cannot reliably know that.

Call it:

> ***Match Score***

and:

> ***Application Priority***

---

73. APPLICATION PRIORITY

Example:

```
Priority =
Eligibility
+
Match
+
Deadline
+
Funding
+
Student Goal
+
Application Readiness
```

```
   APPLY FIRST

94% Match
Deadline: 4 days
Fully funded
Profile: 95% ready
```

```
72% Match
Deadline: 2 months
Partial funding
Profile: 60% ready
```

The first should be prioritized.

74. “OPPORTUNITY RADAR”

Excellent future feature.

```
   STUDENT PROFILE
          │
          ▼
 NEXTPATH MONITOR
          │
          ▼
   NEW OPPORTUNITY
          │
          ▼
    ELIGIBILITY
          │
          ▼
     MATCH SCORE
          │
   ┌──────┴──────┐
   ▼             ▼
 <70%          >80%
   │             │
   ▼             ▼
 Ignore       Notify
                 │
                 ▼
            New opportunity
```

Priority:

Future

Could be simulated in the hackathon.

---

75. PROFILE GAP ANALYSIS

This can become a major future differentiator.

```
Student Profile
       |
       ▼
Opportunities
       |
       ▼
Find near-matches
       |
       ▼
What is preventing eligibility?
       |
       ▼
Skill / document / language / experience
       |
       ▼
Personal roadmap
```

> *Adding IELTS 6.5 could make you eligible for 12 more opportunities.*

This turns NEXTPATH from a **listing platform** into a **student development platform**.

---

76. LONG-TERM PRODUCT

Eventually:

```
                   NEXTPATH
                      │
        ┌─────────────┼─────────────┐
        ▼             ▼             ▼
   DISCOVERY     PREPARATION    APPLICATION
        │             │             │
        ▼             ▼             ▼
 Opportunities   Skill gaps    Autofill
 Matching        Documents     AI writing
 Eligibility     Roadmap       Tracking
        │             │             │
        └─────────────┼─────────────┘
                      ▼
               STUDENT SUCCESS
```

77. THE COMPLETE USER FLOW

```
START
  │
  ▼
Create Account
  │
  ▼
Build Profile
  │
  ▼
Profile Completeness
  │
  ▼
NEXTPATH analyzes profile
  │
  ▼
Retrieve verified opportunities
  │
  ▼
Check Egyptian eligibility
  │
  ▼
Remove impossible opportunities
  │
  ▼
Calculate match
  │
  ▼
Prioritize by deadline + fit
  │
  ▼
Recommend
  │
  ▼
Student opens opportunity
  │
  ├──────────────┐
  ▼              ▼
Eligible      Not Eligible
  │              │
  ▼              ▼
Apply         Why not?
  │              │
  ▼              ▼
Application   Missing requirements
Assist
  │
  ▼
Prepare
  │
  ▼
Review
  │
  ▼
Apply externally
  │
  ▼
Track
  │
  ▼
Deadline reminders
  │
  ▼
RESULT
```

78. HACKATHON DEMO

Scene 1 — Problem

Show:

```
Facebook
Telegram
Google
LinkedIn
University websites
Scholarship websites
```

> *“The problem isn't that opportunities don't exist. The problem is that students don't know which ones are actually meant for them.”*

---

79. Scene 2 — Create student

```
Ahmed
   Egyptian

First-year
Computer Engineering

GPA: 3.5

Skills:
Python
C++
AI

English:
B2

Goals:
Research
Internships
Scholarships
```

80. Scene 3 — AI SEARCH

Type:

> *Find fully funded AI opportunities I can apply for as an Egyptian first-year student.*

NEXTPATH:

```
Searching verified opportunities...

✓ 42 opportunities found
✓ 17 potentially relevant
✓ 9 eligible
✓ 4 high-priority
```

81. Scene 4 — KILLER MOMENT

Open:

94% Match

```
   Egyptian           ✓
First-year             ✓
Computer Engineering   ✓
GPA                    ✓
AI interest            ✓
Age                    ✓

⚠ Recommendation letter
```

> ***“Instead of asking the student to read the entire website, NEXTPATH tells them exactly why this opportunity fits.”***

---

82. Scene 5 — WHY NOT

Open another:

```
   Not eligible

Reason:

Program requires students
to be at least 21 years old.

Your age:
19
```

This demonstrates the intelligence much better than simply filtering it out.

---

83. Scene 6 — APPLICATION

Click:

> *Apply with NEXTPATH*

Show:

```
24 required fields

20 ✓ From profile
2 ⚠ Need confirmation
2    Missing
```

Then AI helps draft the motivation answer.

---

84. Scene 7 — TRACKING

```
MY APPLICATIONS

   AI Research Program
Deadline: 8 days

   Global Hackathon
Submitted

   Scholarship
Missing recommendation
```

85. FINAL DEMO STATEMENT

End with:

> ***“NEXTPATH doesn't just help students find opportunities. It tells them which doors are open, why they are open, what they need to unlock the others, and what they should do next.”***

That's the pitch.

---

86. FINAL FEATURE PRIORITY MATRIX

Feature	Hackathon	Future	Drop
Student profile	·		
Scholarships	·		
Internships	·		
Hackathons	·		
Research	·		
Exchange	·	·	
Training	·		
Events	·		
Volunteer	·		
Egyptian eligibility	·		
Global opportunities	·		
Match score	·		
Why eligible	·		
Why not	·		
AI search	·		
Deadline tracking	·		
Verification	·	·	
Saved opportunities	·	·	
Application tracker		·	
Profile autofill		·	
AI application writing		·	
Opportunity Radar		·	
Skill-gap roadmap			·
Browser auto-apply		·	
Autonomous applications			
Friends/social network			
Native mobile app			

Feature	Hackathon	Future	Drop
Microservices			

87. THE FINAL ARCHITECTURE TO BUILD

If your team gets overwhelmed, come back to this:

```
              NEXTPATH
                 │
                 ▼
         ┌───────────────┐
         │ Student       │
         │ Profile       │
         └───────────────┘
                 │
                 ▼
      ┌────────────────────┐
      │ Verified           │
      │ Opportunities      │
      └────────────────────┘
                │
                ▼
     ┌──────────────────────┐
     │ Eligibility Engine   │
     └──────────────────────┘
              │
         ┌────┴──────┐
         ▼           ▼
      Eligible     Not eligible
         │             │
         ▼             ▼
   Match Engine     WHY NOT?
         │
         ▼
   Match Score
         │
         ▼
   AI Explanation
         │
         ▼
   Recommendation
         │
         ▼
 Application Assistant
         │
         ▼
   Student Review
         │
         ▼
  External Application
         │
         ▼
   Application Tracker
         │
         ▼
     Notifications
```

The three things your judges should remember

1. DISCOVER

> *NEXTPATH finds opportunities globally.*

2. UNDERSTAND

> *NEXTPATH tells Egyptian students whether they can actually apply — and why.*

3. ACT

> *NEXTPATH turns a matching opportunity into a prepared, trackable application.*

That is the **real NEXTPATH product**. The AI is the intelligence layer—not the product itself. The product is the entire pipeline from **“I don't know what's out there”** → **“I know exactly what I should apply for next.”**